



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI VENKATESHWARA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

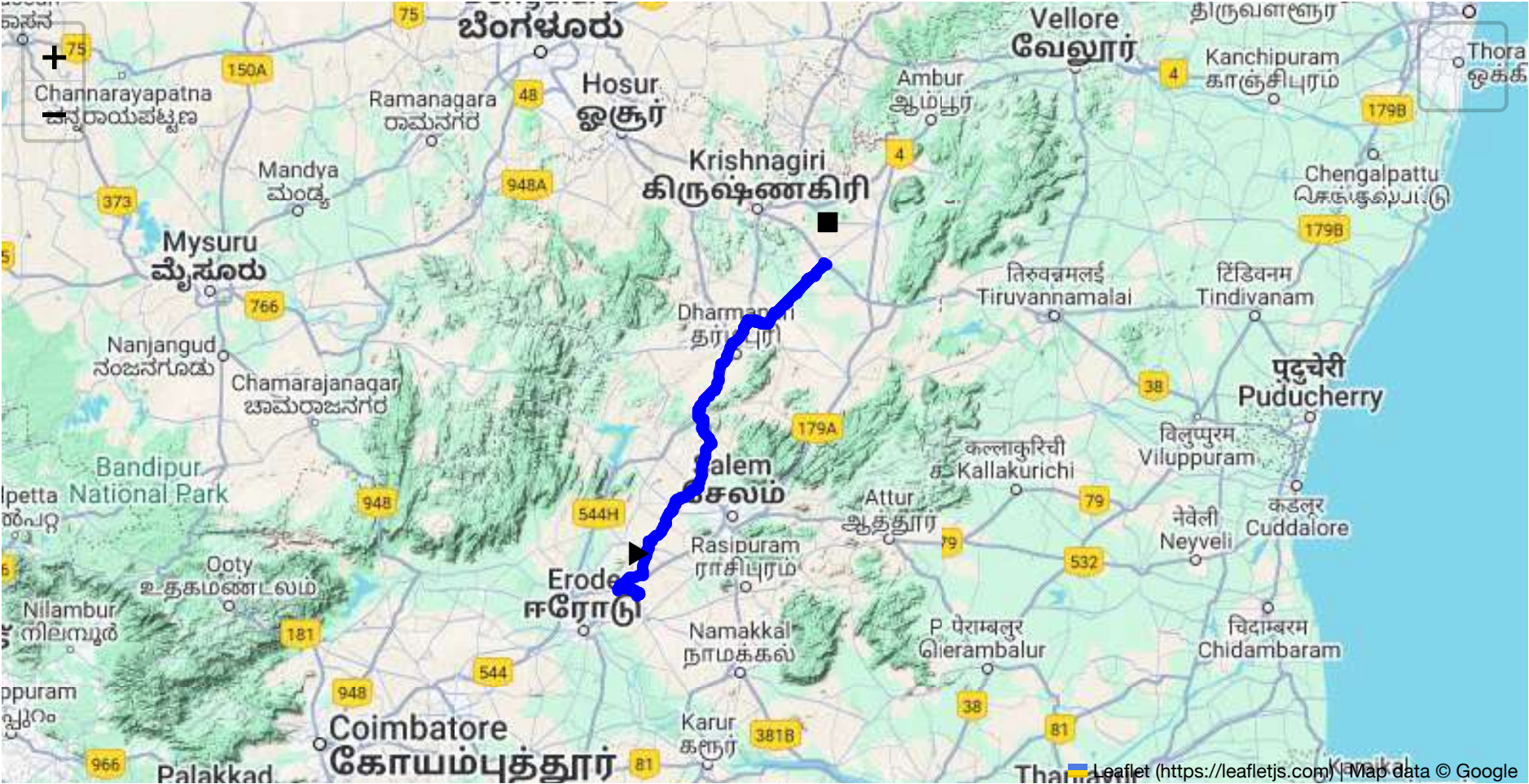
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 149.53 km
Estimated Duration: 3.2 hours
Adjusted Duration (Heavy Vehicle): 4.0 hours

Start: (11.4381, 77.8734)
End: (12.370864, 78.415576)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The journey spans approximately 149.53 kilometers from Sangagiri to Batrahalli in Tamil Nadu, India. The route includes several waypoints:

- FR48+7XQ, Padaiveedu
- FVVP+J73, Ayveli
- HW72+H83, Erumaipatti
- P2PM+M9M, Salem
- 665W+4R9, Krishnapuram

This combination of state highways and local roads weaves through varied terrain, including urban, semi-urban, and rural areas.

2. Typical Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate.

- Summer (March to June): High temperatures can affect vehicle performance.
- Monsoon (July to September): Heavy rainfall can lead to waterlogging and decreased visibility.
- Winter (November to February): Mild conditions, but fog might occur in some areas.

Potential weather hazards include cyclones and thunderstorms, especially during the monsoon season.

3. Traffic Patterns

- Peak Hours: Morning (7-9 AM) and evening (5-8 PM), particularly around Salem and major highways.

- **Congestion-prone Areas:** Salem is notably busy, along with other towns like Dharmapuri. Expect slow traffic near marketplaces and urban centers.

4. Road Quality and Infrastructure

- **State Highways (e.g., SH 86):** Generally well-maintained but watch for potholes during monsoon.
- **Local Roads:** Variable quality, with potential for narrow lanes and sudden turns.
- **Infrastructure:** Adequate signage in urban areas, but less frequent in rural sections.

5. Alternative Routes for Emergencies

- **Diversion through NH 44:** If congestion or hazards occur near Salem, NH 44 offers a parallel pathway though longer.
- **Local Bypasses:** Use local knowledge to access smaller bypass roads around towns.

6. Summary of Local Regulations

Tamil Nadu follows national regulations regarding hazardous material transport, requiring:

- Proper labeling and documentation.
- Following specified travel timings to avoid peak hours.

Fines or penalties might levy for non-compliance.

7. Historical Incidents

Records indicate few major incidents involving heavy vehicles along SH 86, but minor accidents attributed to speeding and overtaking have occurred, particularly near urban centers like Salem.

8. Environmental Considerations

- **Sensitive Areas:** Avoid discharge near water bodies, farms, and populated areas.
- **Preservation Zones:** Watch for protected wildlife areas, though these are limited along the route.

9. Communication Coverage

- **Urban Areas:** Strong mobile network coverage, including 4G.
- **Rural Areas:** Possible dead zones; advisable to plan communication points ahead.

10. Estimated Emergency Response Times

- **Urban Areas (Salem):** Approximately 15-30 minutes.
- **Rural Areas:** Can range from 30 minutes to over an hour, depending on proximity to emergency services.

11. Overall Summary of Risk Assessment

The route is generally safe but requires caution, particularly in terms of weather-related risks and traffic congestion near urban areas. Road quality varies, necessitating observance of speed limits and attentive driving. Compliance with hazardous material transport regulations is essential. Ensuring alternate

communication methods is advised in less-connected segments. Emergency services are more accessible in urban centers; hence, planning stops accordingly is prudent.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	Medium	11.45546, 77.81800	30 KM/Hr
7	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
8	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
9	Turn	High	11.49469, 77.88488	15 KM/Hr
10	Turn	Medium	11.56072, 77.89919	30 KM/Hr
11	Turn	Medium	11.56140, 77.89991	30 KM/Hr
12	Turn	Medium	11.59085, 77.90592	30 KM/Hr
13	Turn	Medium	11.59253, 77.90690	30 KM/Hr
14	Turn	Medium	11.63848, 77.94825	30 KM/Hr
15	Turn	High	11.64197, 77.95197	15 KM/Hr
16	Turn	Medium	11.64558, 77.95108	30 KM/Hr
17	Turn	Medium	11.67883, 77.96865	30 KM/Hr
18	Turn	High	11.70227, 77.97864	15 KM/Hr
19	Turn	Medium	11.71386, 77.99552	30 KM/Hr
20	Turn	Medium	11.71735, 78.00019	30 KM/Hr
21	Turn	Medium	11.72765, 78.02291	30 KM/Hr
22	Turn	High	11.73785, 78.03496	15 KM/Hr
23	Turn	High	11.73653, 78.03886	15 KM/Hr
24	Turn	Medium	11.74252, 78.04705	30 KM/Hr
25	Turn	Medium	11.77998, 78.05786	30 KM/Hr
26	Turn	Medium	11.86915, 78.08382	30 KM/Hr
27	Turn	Medium	11.87998, 78.08170	30 KM/Hr
28	Turn	Medium	11.91266, 78.05852	30 KM/Hr
29	Turn	Medium	11.92653, 78.06517	30 KM/Hr
30	Turn	Medium	11.93196, 78.06473	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
31	Turn	Medium	11.93473, 78.06268	30 KM/Hr
32	Turn	Medium	11.93746, 78.05331	30 KM/Hr
33	Turn	Medium	11.95531, 78.04698	30 KM/Hr
34	Turn	High	11.95854, 78.04990	15 KM/Hr
35	Turn	High	11.96015, 78.04945	15 KM/Hr
36	Turn	Medium	11.96266, 78.05115	30 KM/Hr
37	Turn	High	11.96433, 78.05113	15 KM/Hr
38	Turn	High	11.96542, 78.04938	15 KM/Hr
39	Turn	High	11.96650, 78.04927	15 KM/Hr
40	Turn	Medium	12.17041, 78.16823	30 KM/Hr
41	Turn	Medium	12.21851, 78.19160	30 KM/Hr
42	Turn	Medium	12.21883, 78.19247	30 KM/Hr
43	Turn	Medium	12.21619, 78.19634	30 KM/Hr
44	Turn	Medium	12.21373, 78.21093	30 KM/Hr
45	Turn	High	12.21454, 78.21229	15 KM/Hr
46	Turn	High	12.21200, 78.21640	15 KM/Hr
47	Turn	Medium	12.21232, 78.21749	30 KM/Hr
48	Turn	High	12.21131, 78.22316	15 KM/Hr
49	Turn	Medium	12.21082, 78.22342	30 KM/Hr
50	Turn	High	12.20520, 78.24481	15 KM/Hr
51	Turn	Medium	12.26781, 78.30424	30 KM/Hr
52	Turn	Medium	12.26949, 78.30696	30 KM/Hr
53	Turn	Medium	12.28421, 78.32011	30 KM/Hr
54	Turn	Medium	12.28508, 78.32270	30 KM/Hr
55	Turn	Medium	12.34223, 78.38402	30 KM/Hr
56	Turn	High	12.37522, 78.40967	15 KM/Hr
57	Turn	Medium	12.37480, 78.41230	30 KM/Hr
58	Turn	High	12.37125, 78.41567	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital, Pochampalli	12.3322477, 78.365767	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
31	school	GUNA Matric Higher Secondary School	12.3722333, 78.4086433	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



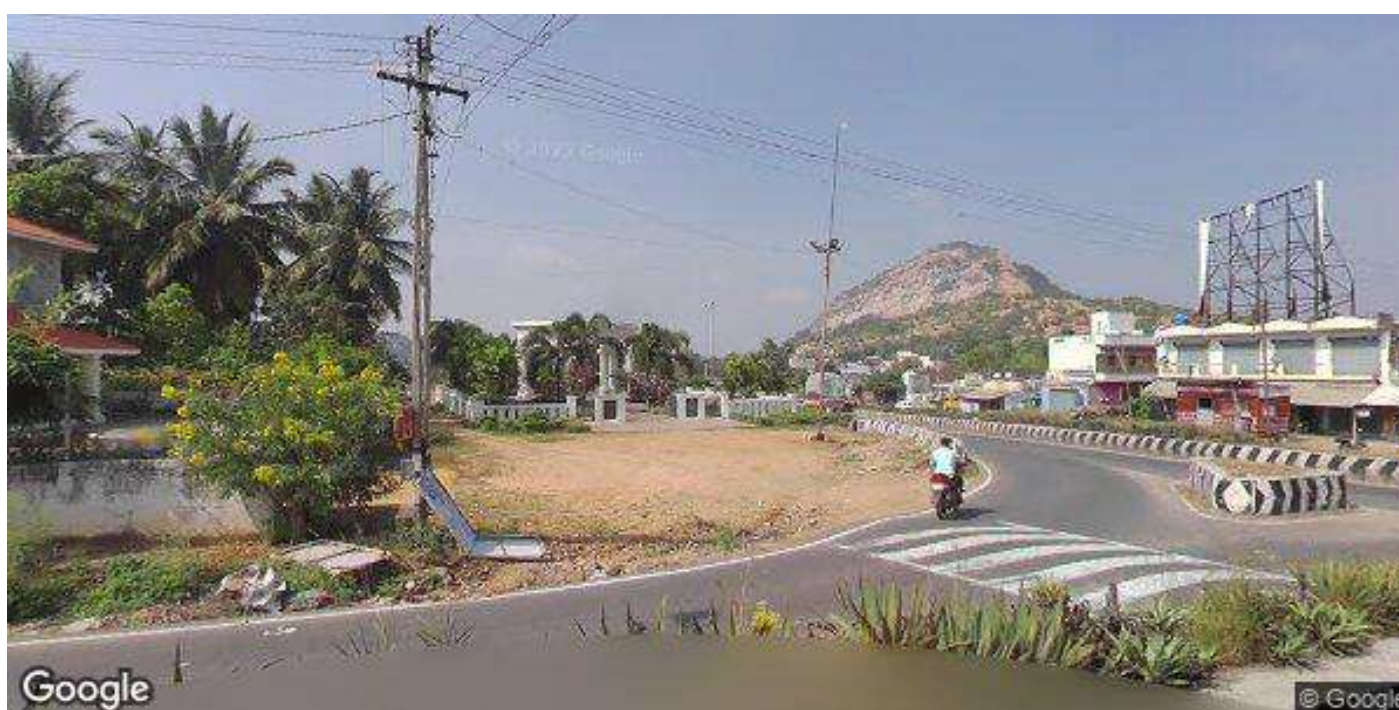
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56072, 77.89919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56140, 77.89991



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59085, 77.90592



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59253, 77.90690



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63848, 77.94825



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64558, 77.95108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67883, 77.96865



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73785, 78.03496



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73653, 78.03886



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74252, 78.04705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87998, 78.08170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92653, 78.06517



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93196, 78.06473



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93473, 78.06268



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96266, 78.05115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17041, 78.16823



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21851, 78.19160



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21883, 78.19247



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21619, 78.19634



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21373, 78.21093



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21454, 78.21229



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21200, 78.21640



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21131, 78.22316



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21082, 78.22342



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.20520, 78.24481



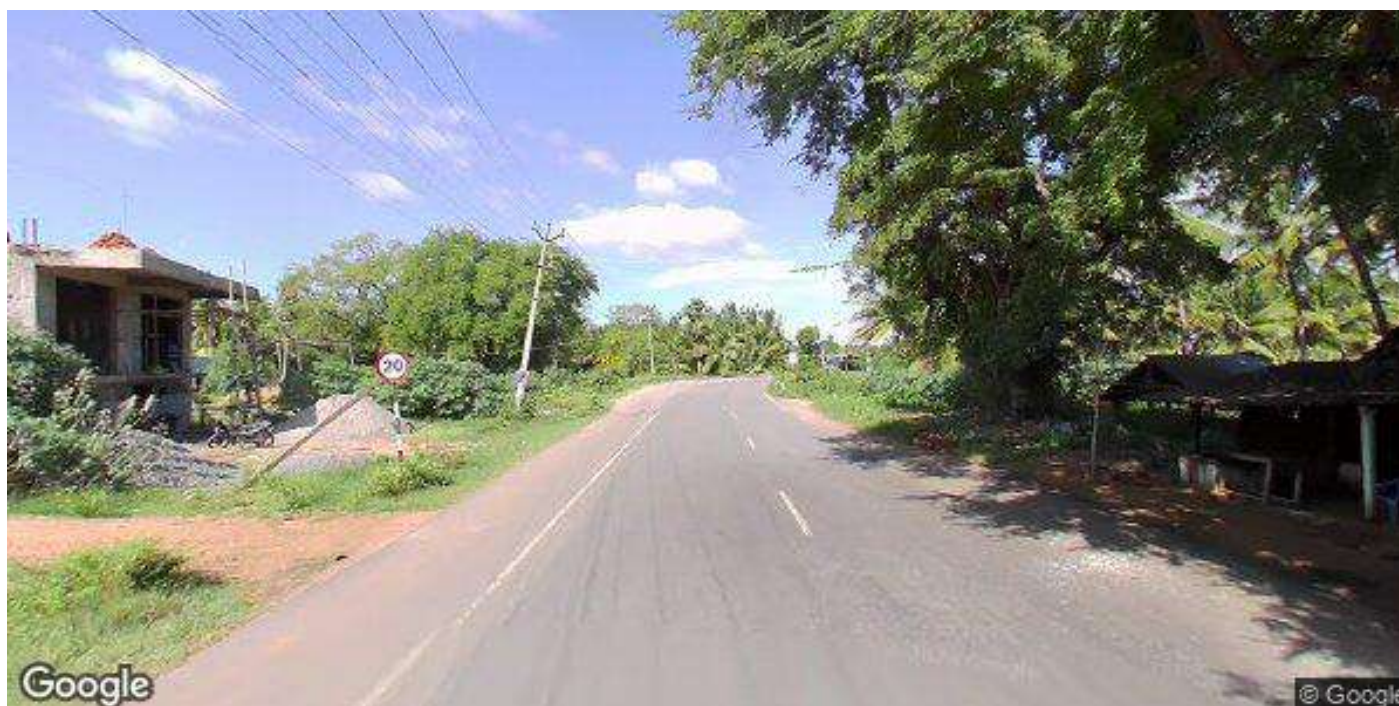
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26781, 78.30424



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26949, 78.30696



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.28421, 78.32011



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.28508, 78.32270



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.34223, 78.38402



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.37522, 78.40967



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.37480, 78.41230



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.37125, 78.41567
